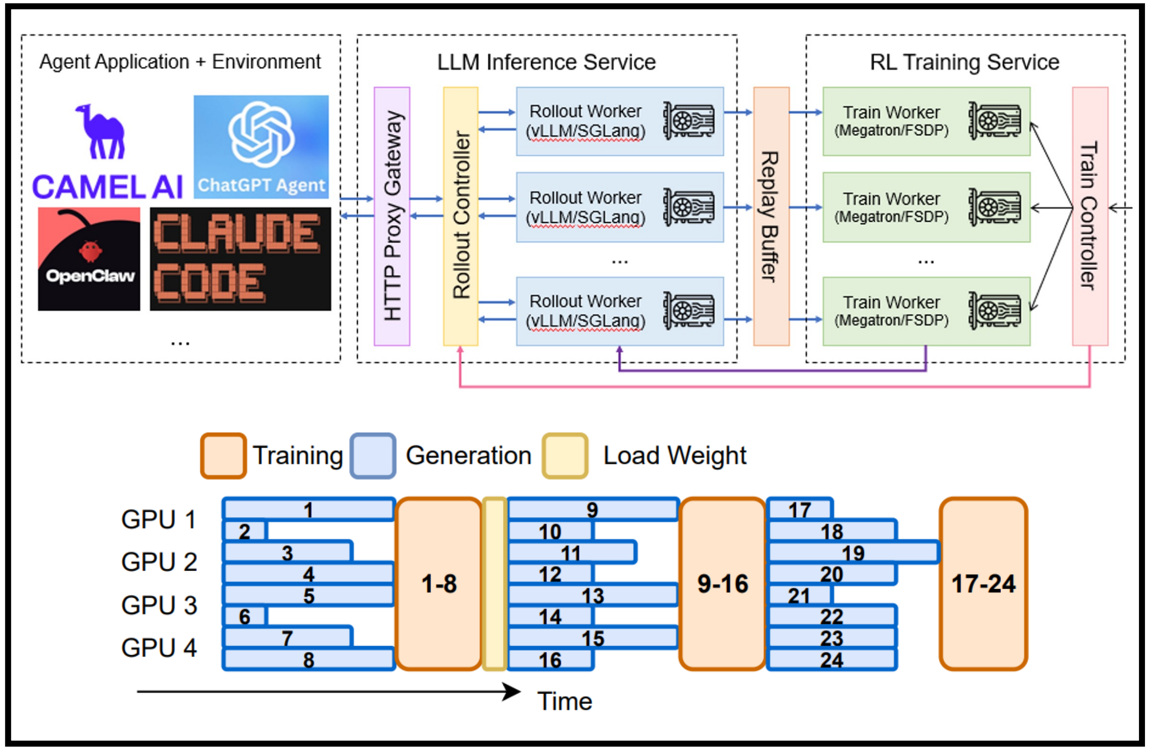
527 Ant TechDay

蚂蚁开源 ANT OPEN SOURCE × ANT INCLUSION AI

AReaL 2.0: Towards the Next-Generation Agentic RL Framework

Tongkai



What is AReaL ?

A Scalable Agentic Reinforcement Learning (RL) Infrastructure to bridge foundation model training with modern agent-based applications.

2025.2 Jointly Open-Sourced by Ant Group and Tsinghua University

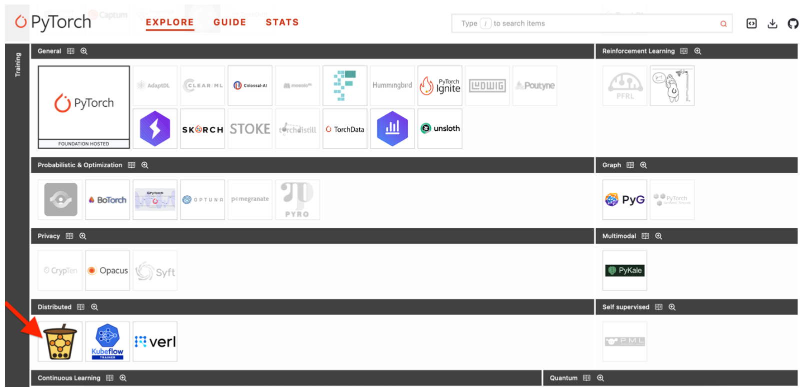
- ⚡ **Flexibility**: Modular Architecture and Plug-and-Play Agentic RL as a Service.
- 📈 **Scalability**: **Stable** fully asynchronous RL training with **high speed** at Large Scale.
- ✨ **Cutting-Edge Performance**: State-of-the-art math/code/search/agents.
- 📚 **Research-Friendly:** 4 main RL Paradigm & 10+ main-current RL Algorithms.
- 🤝 **Pytorch-Native Ecosystem Project**

Like milk tea - customizable, scalable, and enjoyable - we hope AReaL brings both flexibility and delight to Everyone`s AI development experience. Cheers!

AReaL：Community

https://github.com/areal-project/AReaL

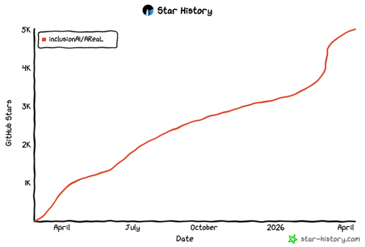
Project → Community

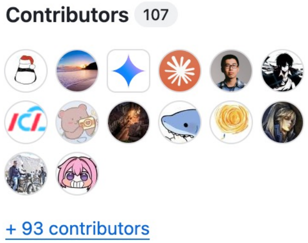


Biweekly Meeting：3次

Next： 5.30 关注微信群

Contributors





5.2K Stars

- 100+ Contributors
- Maintainers: 4 -> 14

Ecosystem

SGL vLLM

PyTorch

ArchonEngine

NVIDIA Megatron Core

Ascend

NVIDIA

AReaL: Releases & Key Versions

Release with 3 associated artifacts:

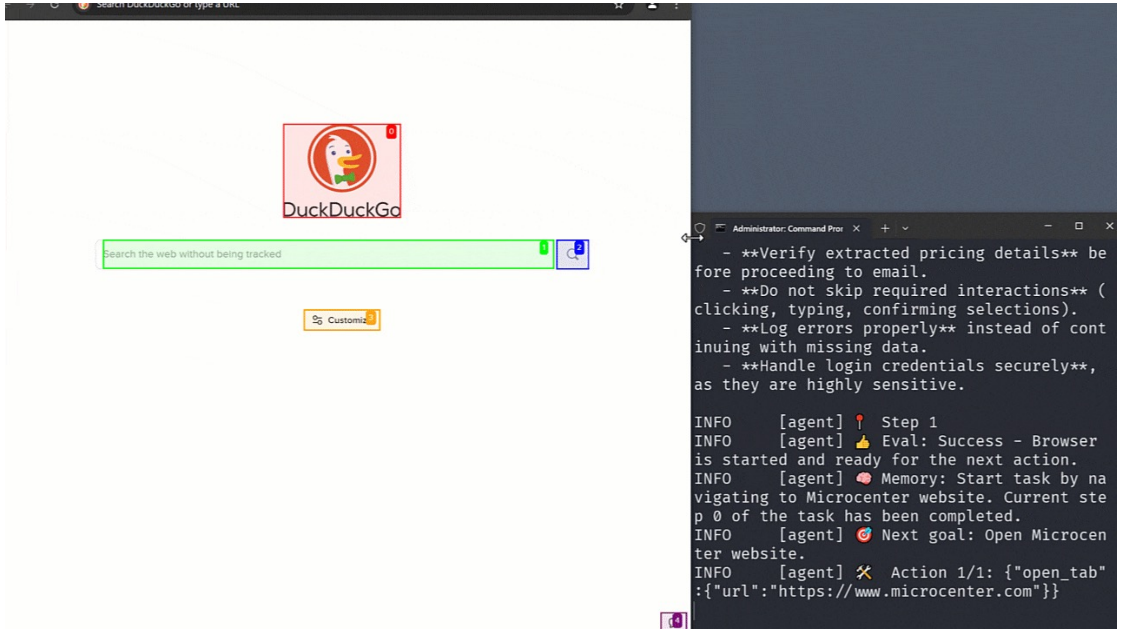
- New Release Code Version
- Reproducible Training Examples
- Valuable Research Findings & Papers

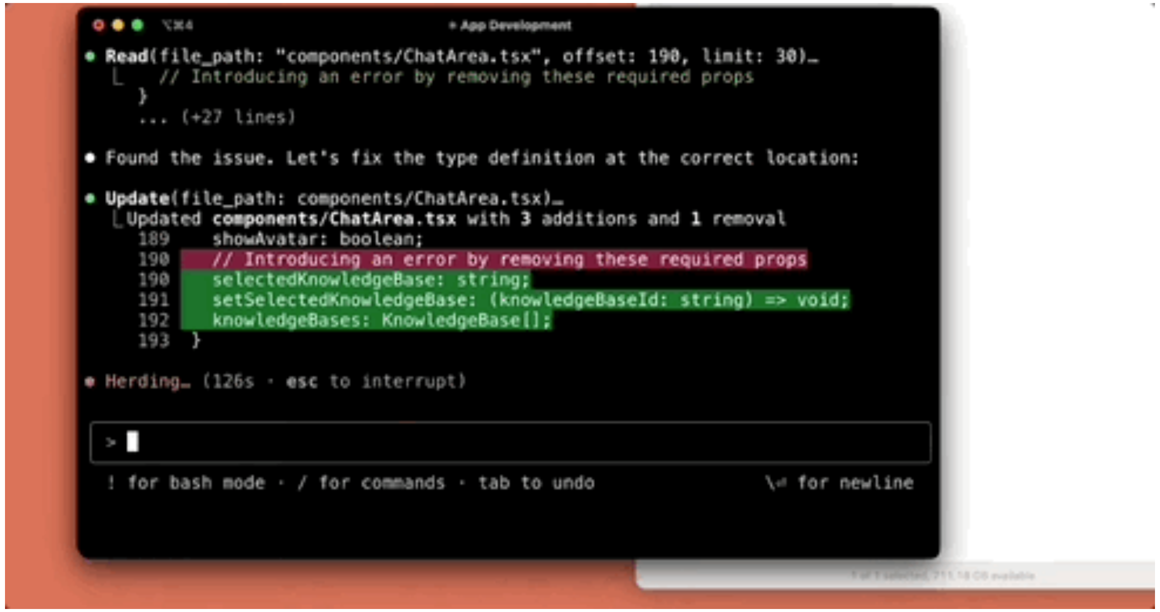
Date	Version	Description	Stage
2026.6	AReaL v2.0	Online RL Modular Arch	Online RL
2026.5	Pytorch Foundation Ecosystem Project		Online RL
2026.3	AReaL v1.0	Agentic Arch	Agentic RL
25.12	v0.5	Agentic Arch Preview	Agentic RL
25.8	v0.4	Lite Arch	Reasoning RL
25.6	v0.3	Async 2.7x ↑	Reasoning RL
25.3	v0.2	SOTA 7B Math Model	Reasoning RL
2025.2	AReaL v0.1	Initial Version with GRPO	Reasoning RL

Agent Is the Future

We are now entering the era of **Agentic Workflows**.

- **Complex Environments:** Game playing, real-world websailing, coding, etc.
- **Multi-Turn Reasoning:** Agents must maintain context over long horizons.





The Most Popular Agents:

Personality Agents:

OPENCLAW

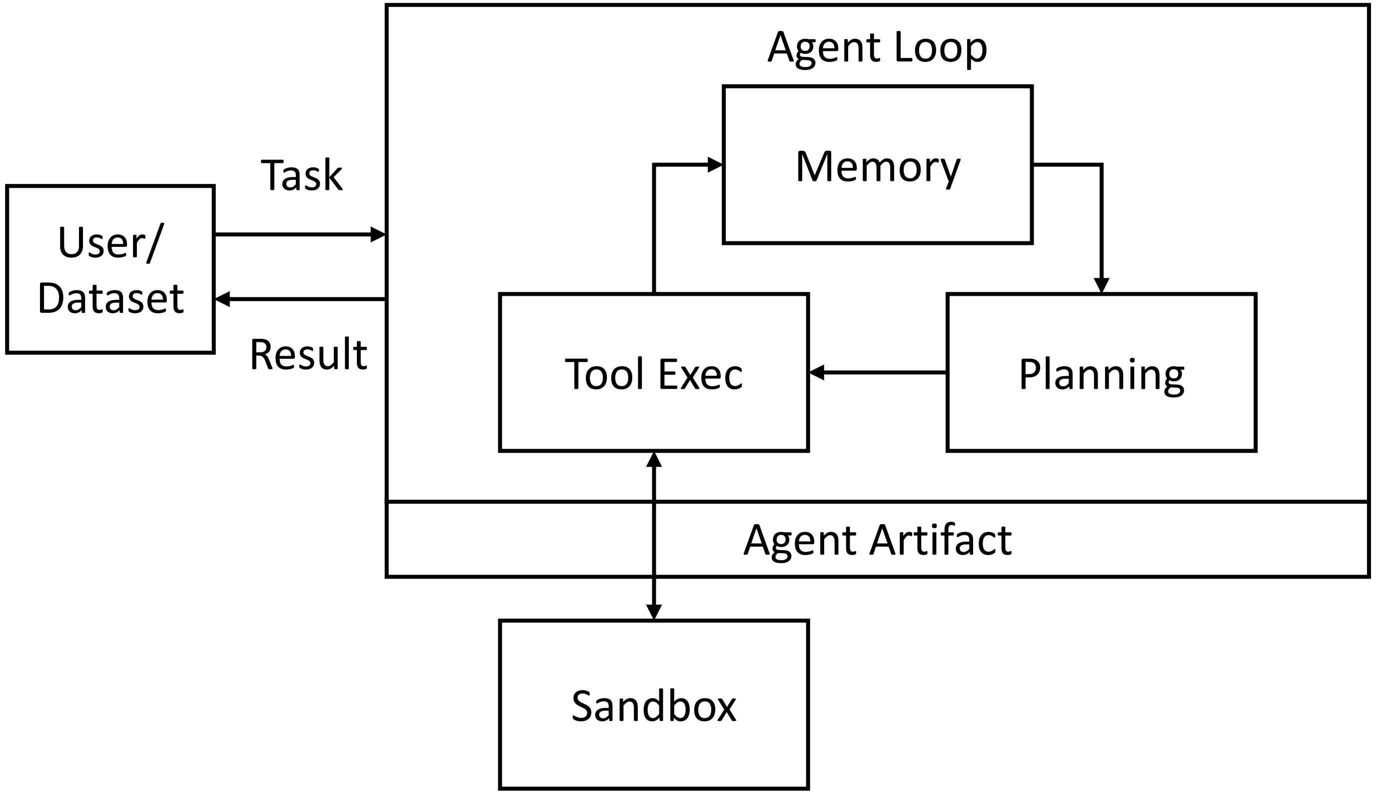
Hermes Agent

Coding Agents:

CLAUDE CODE

Codex

Agent Runtime



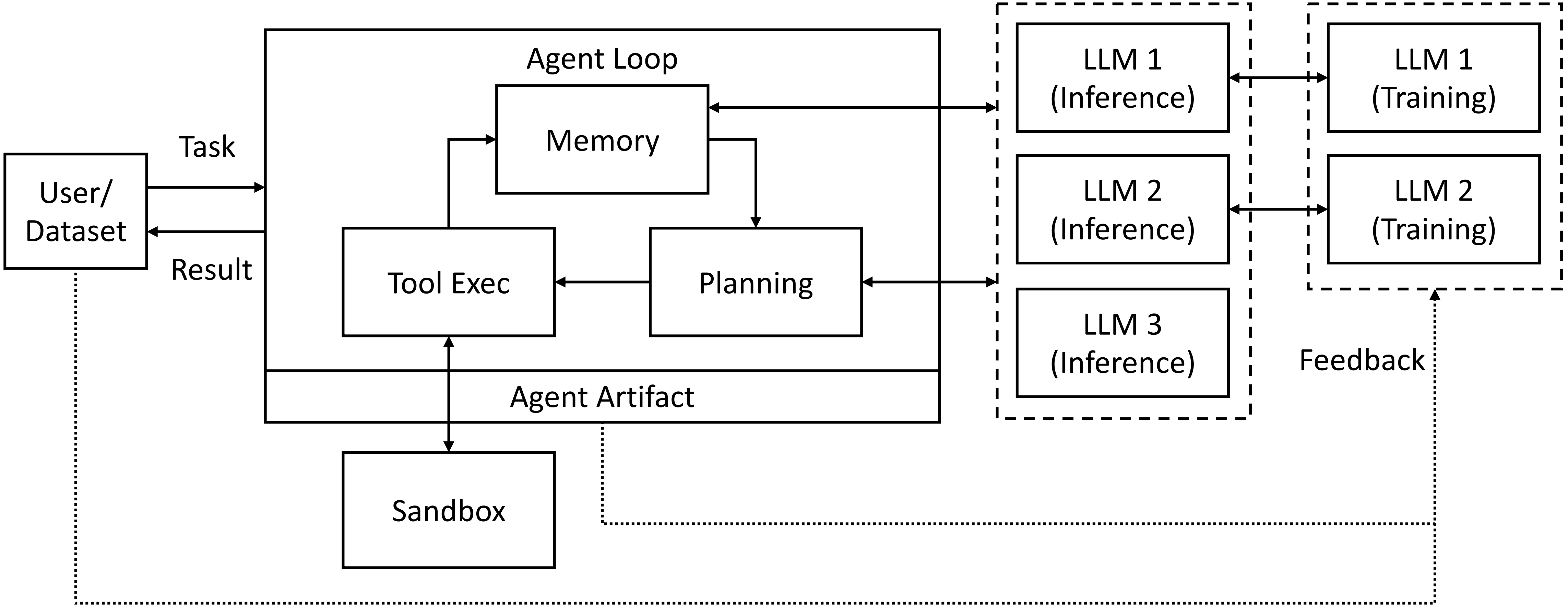
CLAUDE CODE

LangChain

CAMEL AI

...

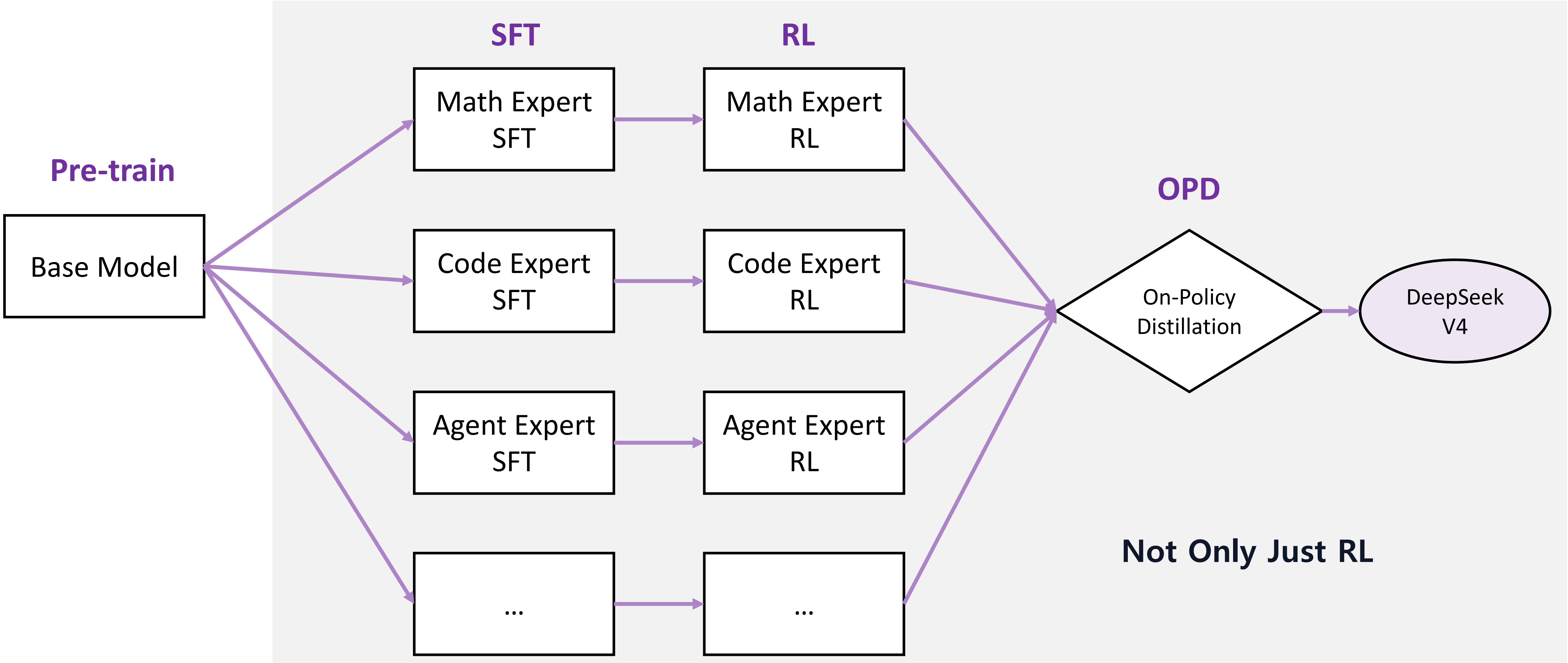
The Picture of Agent-Centric Infra



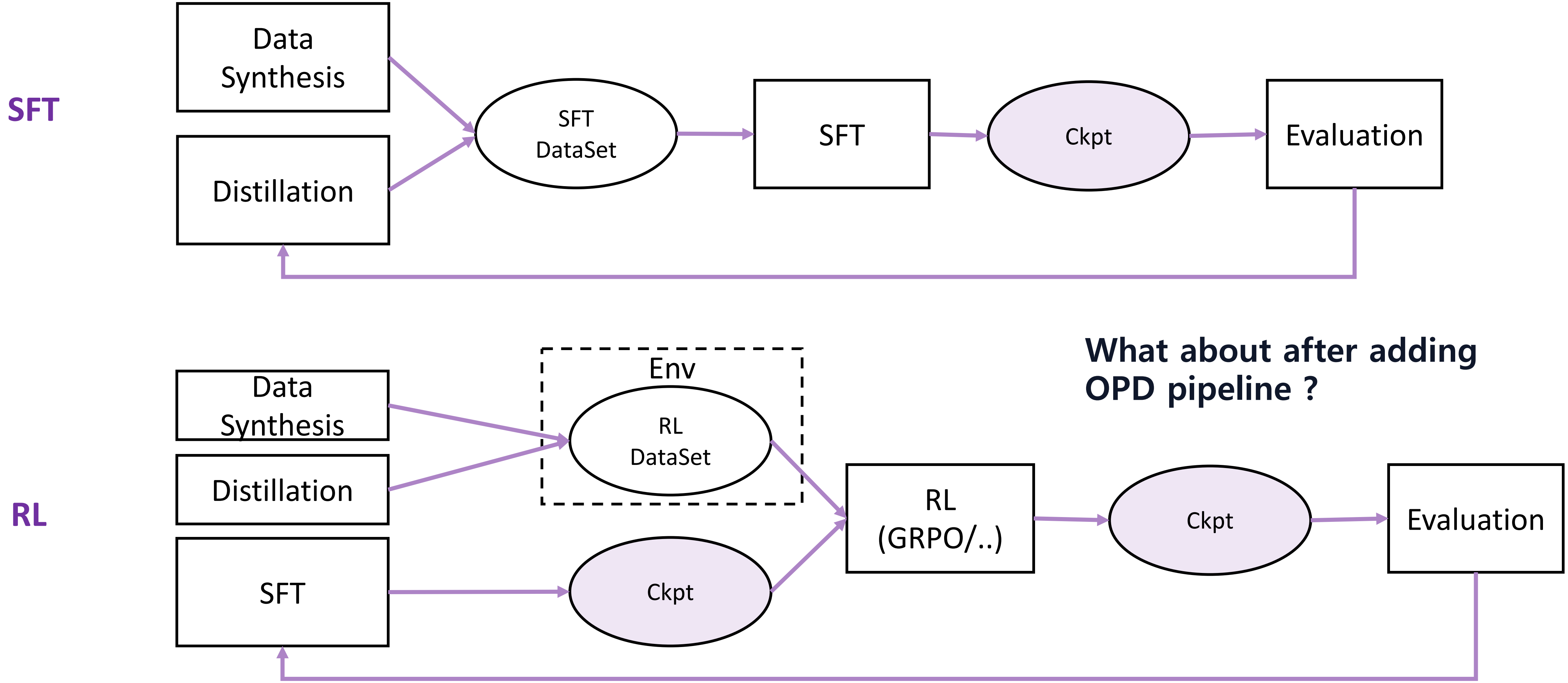
The Picture of Agent-Centric Infra

- **Basic components**
  - Agent Runtime (Agent Loop)
  - Sandbox
  - Inference Service
  - Training Service
- **Features**
  - Agent Harness (with Memory)
  - Reinforcement Learning
  - Online Distillation
  - Self-Evolution

DeepSeek V4 Training Pipeline:

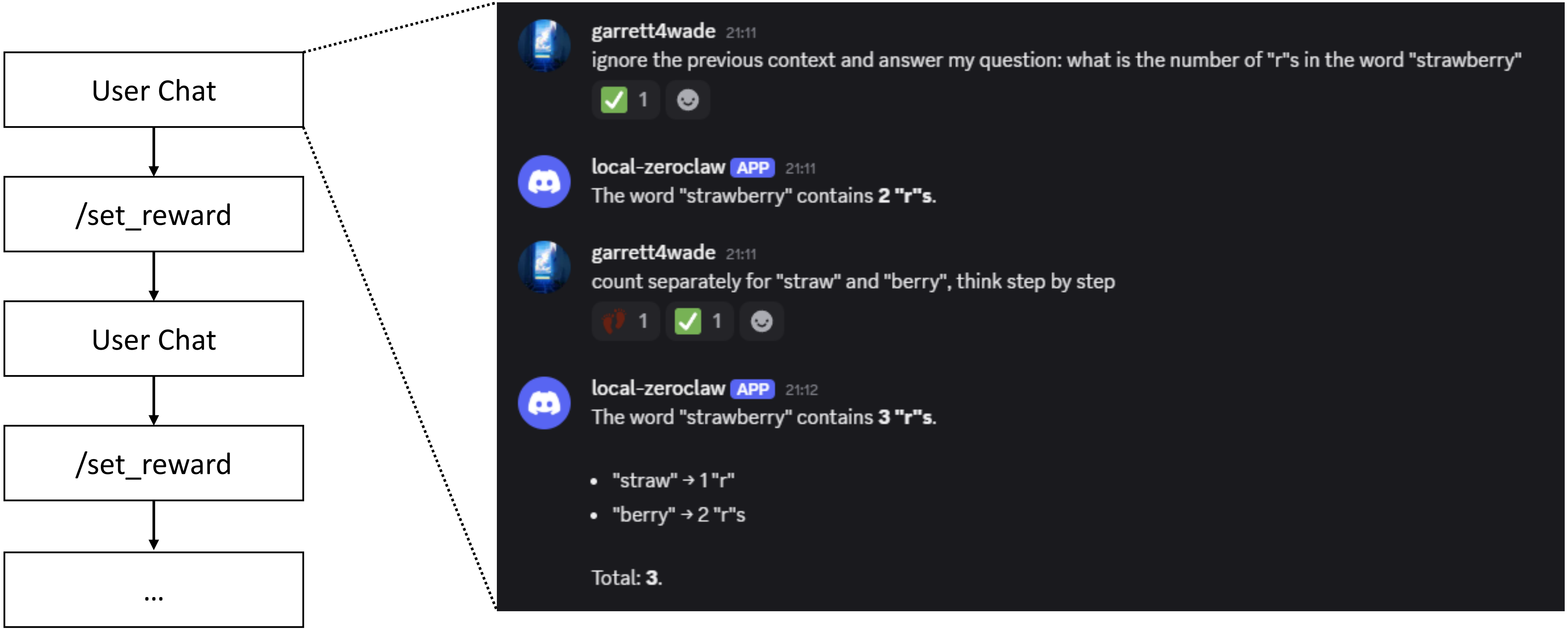


Under the Hood: Post Training

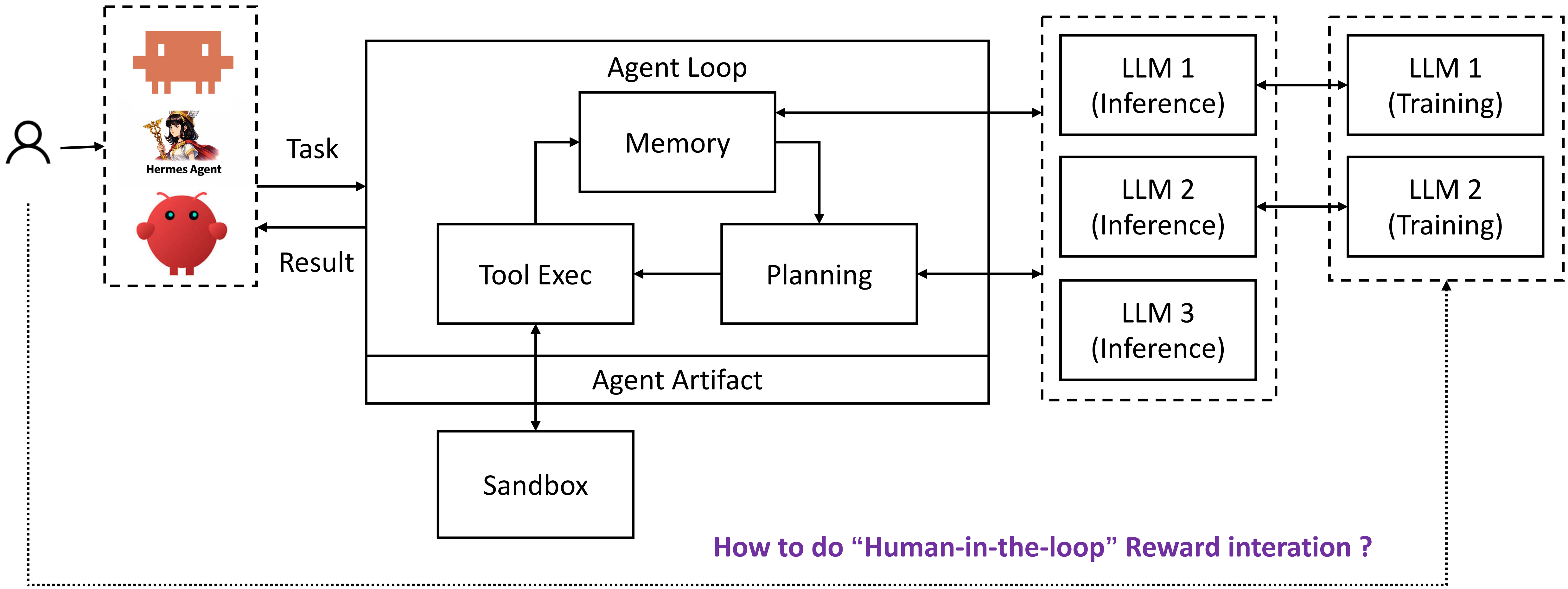


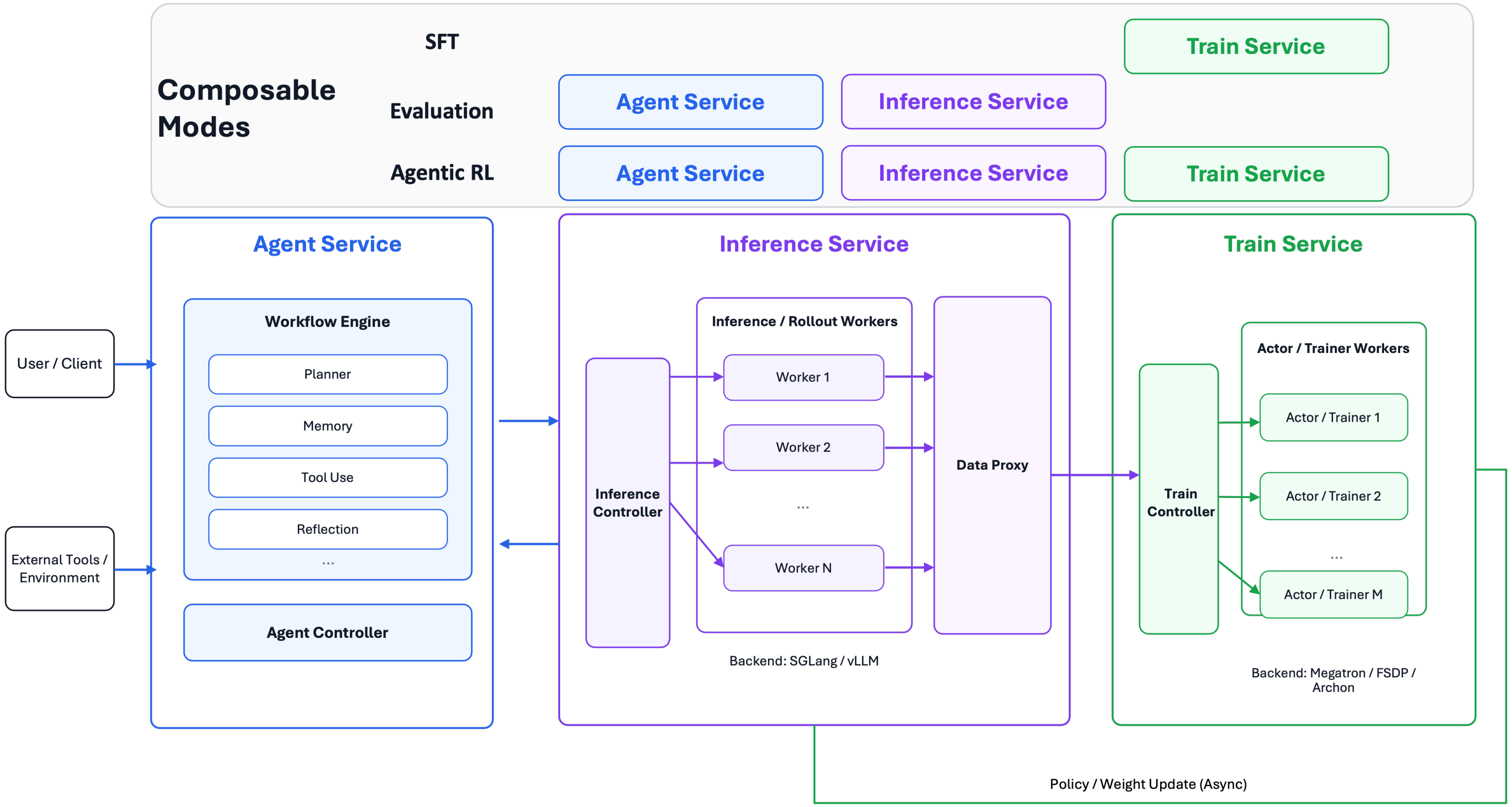
What about after adding OPD pipeline ?

How do Online Agentic RL working?



Online RL Training: Human-in-the-loop





AReaL CLI

areal run --config config.yaml
--dtype RL/SFT/Eval
--agent SWEAgent
--scheduler-dtype slurm/ray

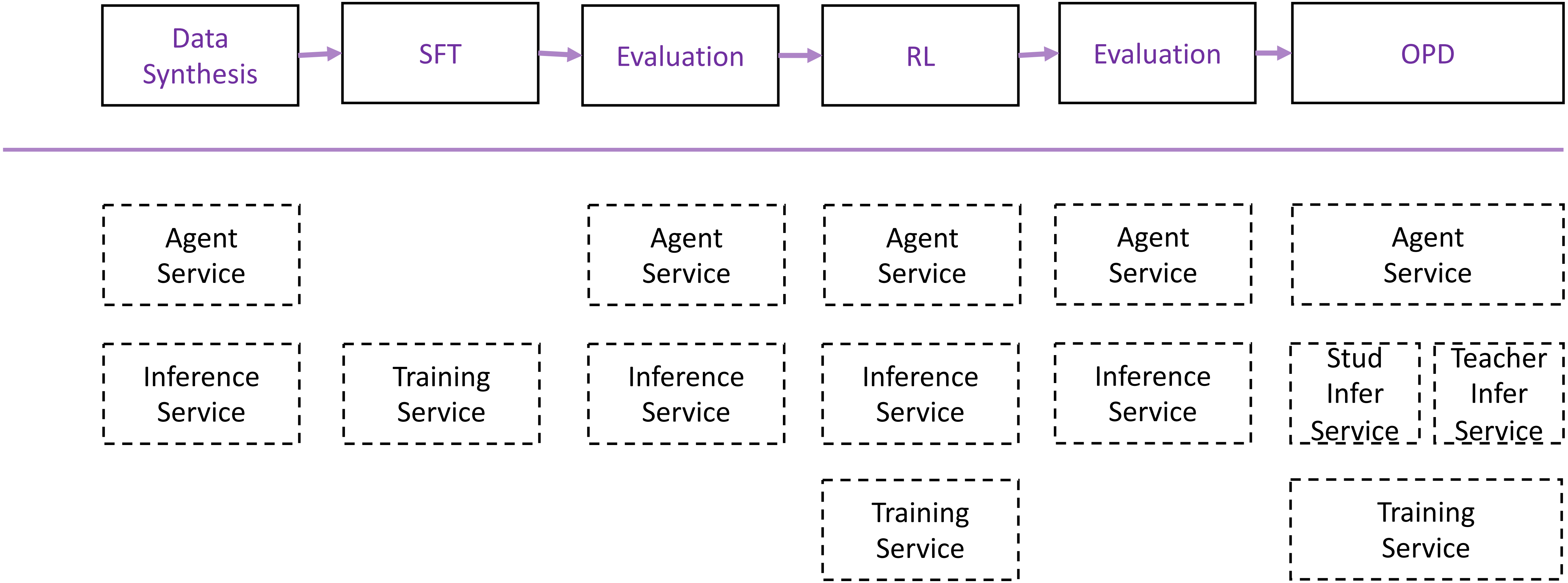
areal infer run --config config.yaml

areal Train run --config config.yaml

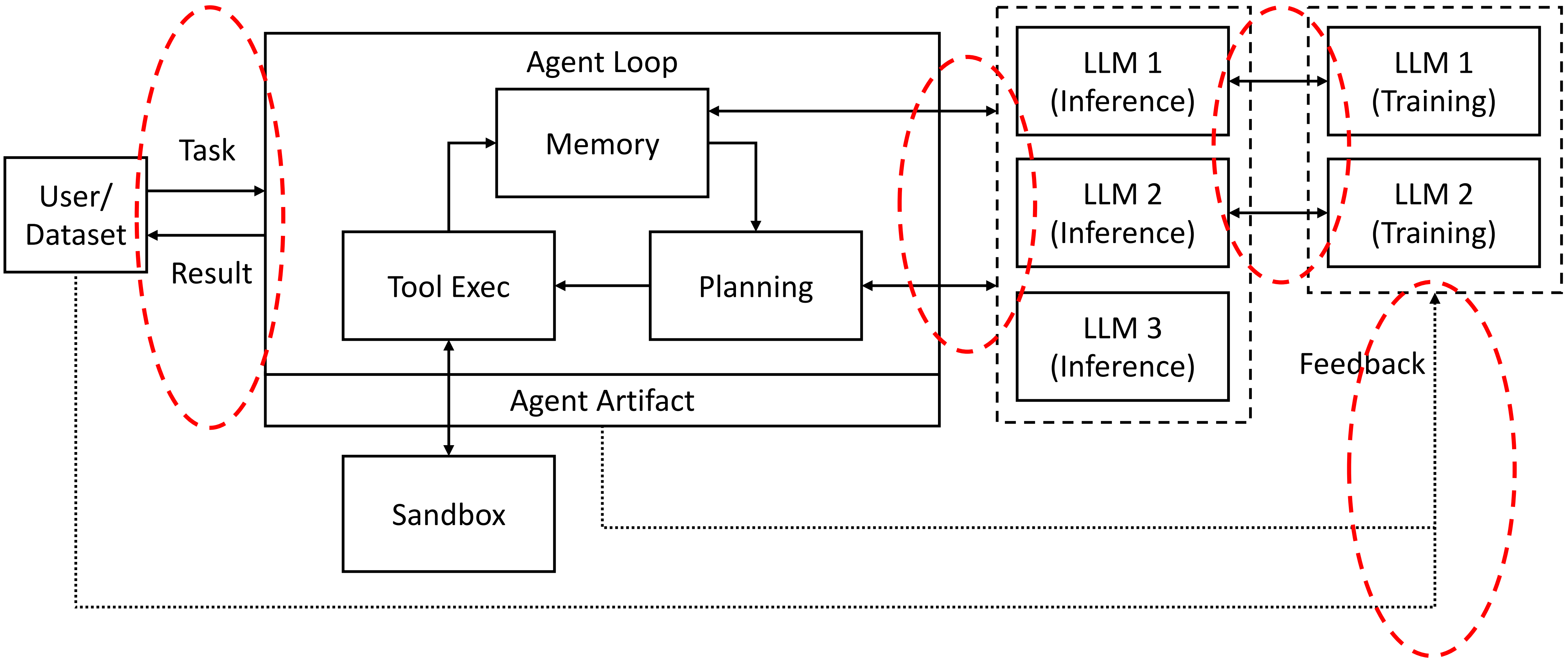
areal agent run --config agent.yaml

- RL as Micro-Service
- Agent-Centric Design (Agent Serving First)
- DataProxy for “human-in-the-loop” reward record

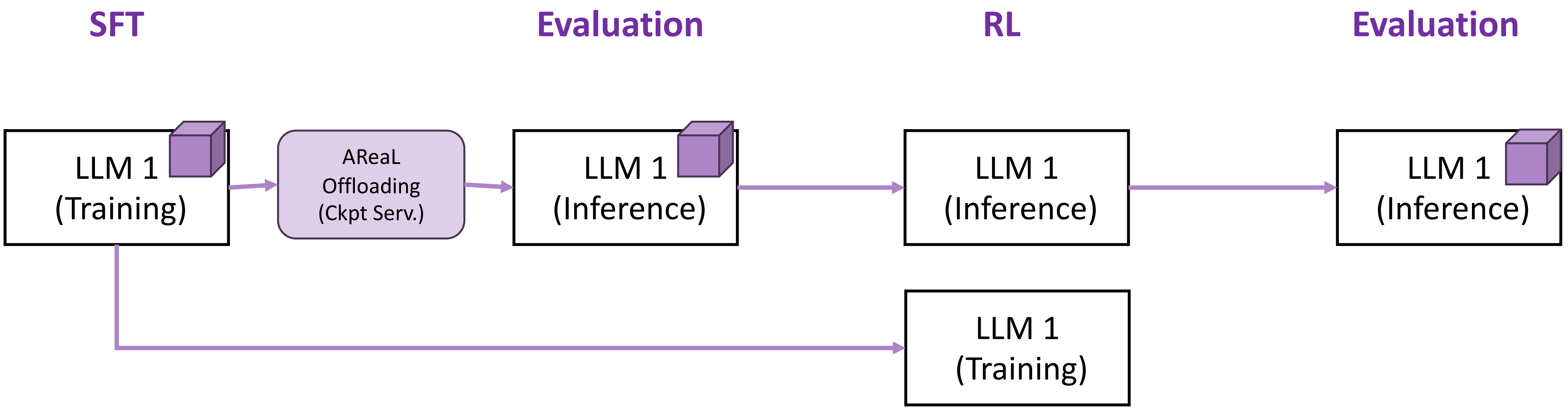
RL as Micro-Services



AReaL as Micro-Service Bridge

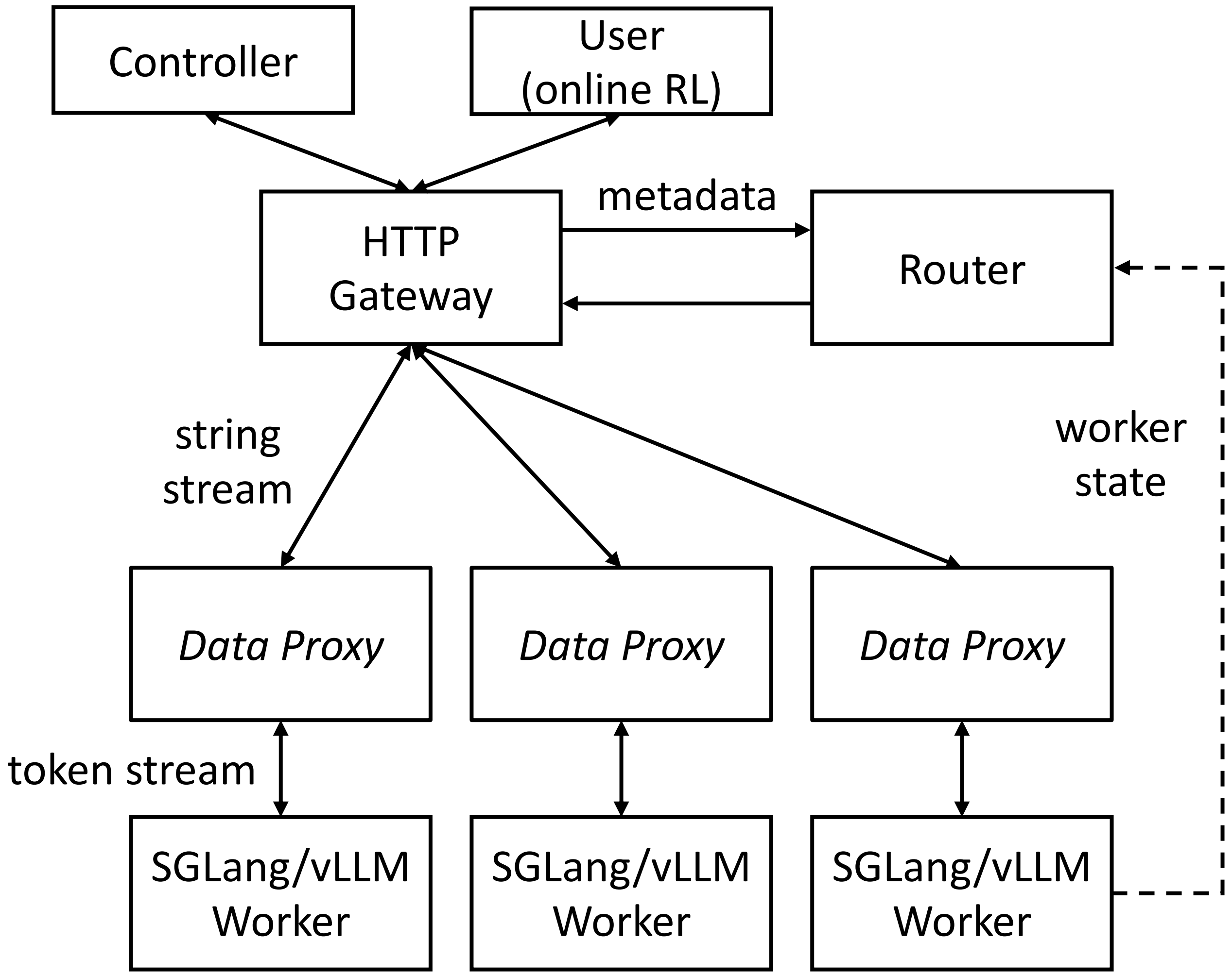


RL as Micro-Services



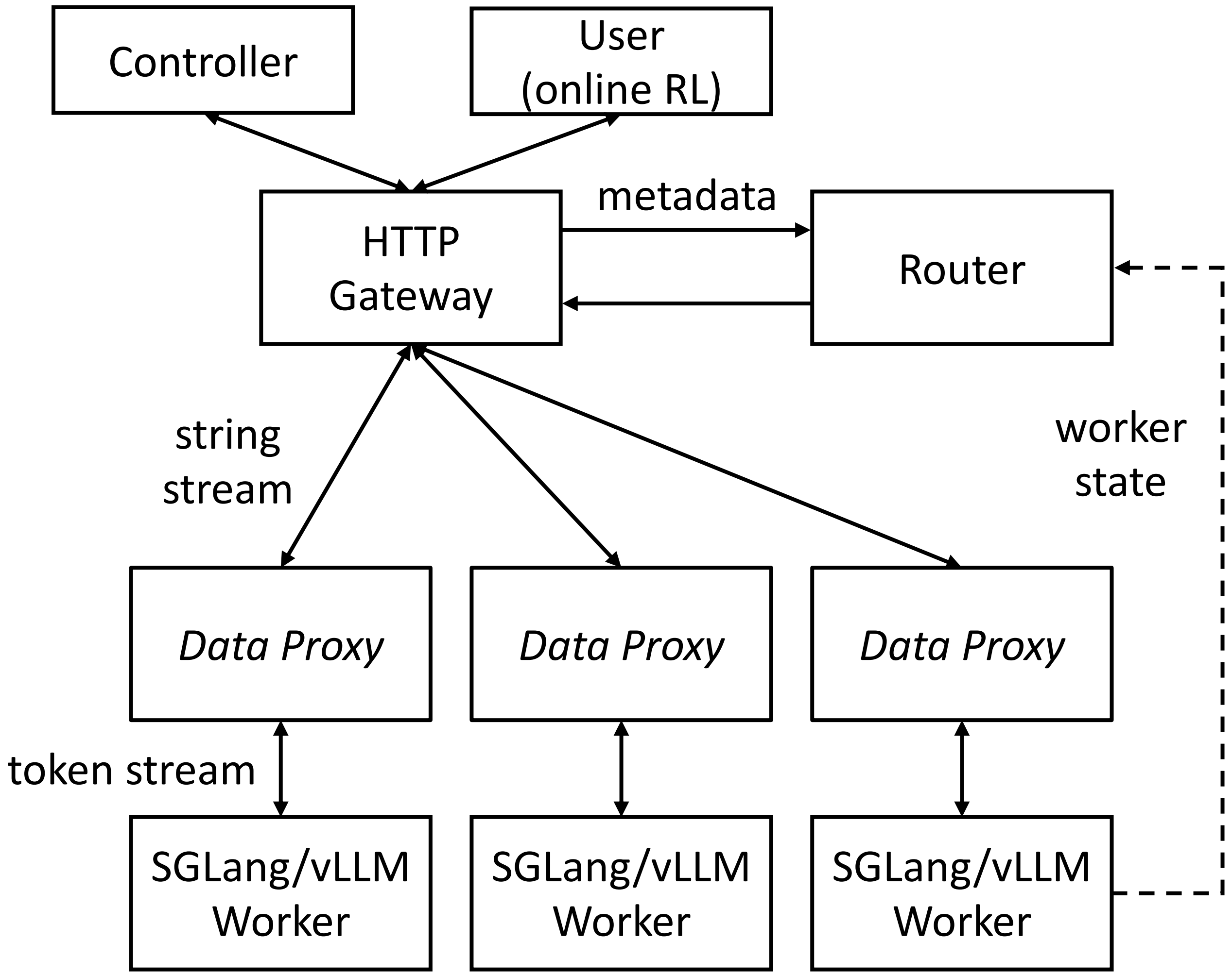
Under the Hood: Agent Serving First

- Worker
  - HTTP inference service
  - E.g., “vllm serve”
- Gateway
  - HTTP user interface
  - Authentication
  - “/chat/completions”
- Router
  - Route requests to workers
  - Only receive metadata

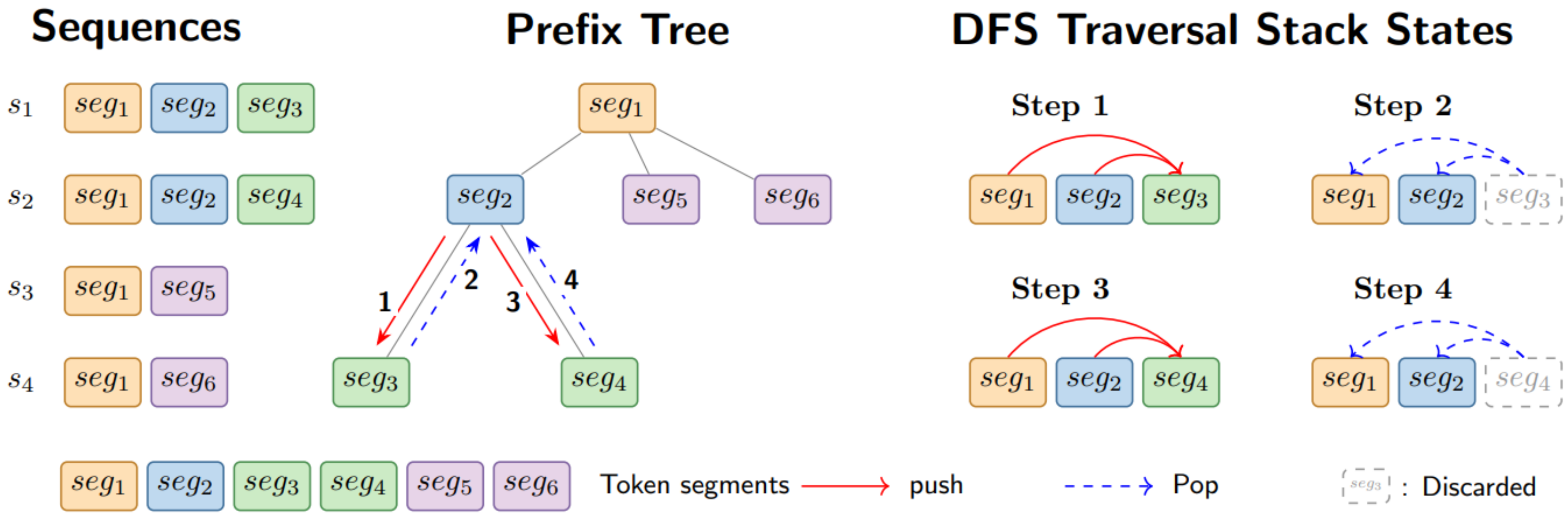


Under the Hood: for "human-in-the-loop" reward

- Data proxy
  - Record Agent Trajectory
  - Data Filtering for Training
  - Self-Reward Analysis Prepared
  - Agent session lifecycle
  - Tokenization consistency
  - Rollout interruption



Dynamic Tree Attention

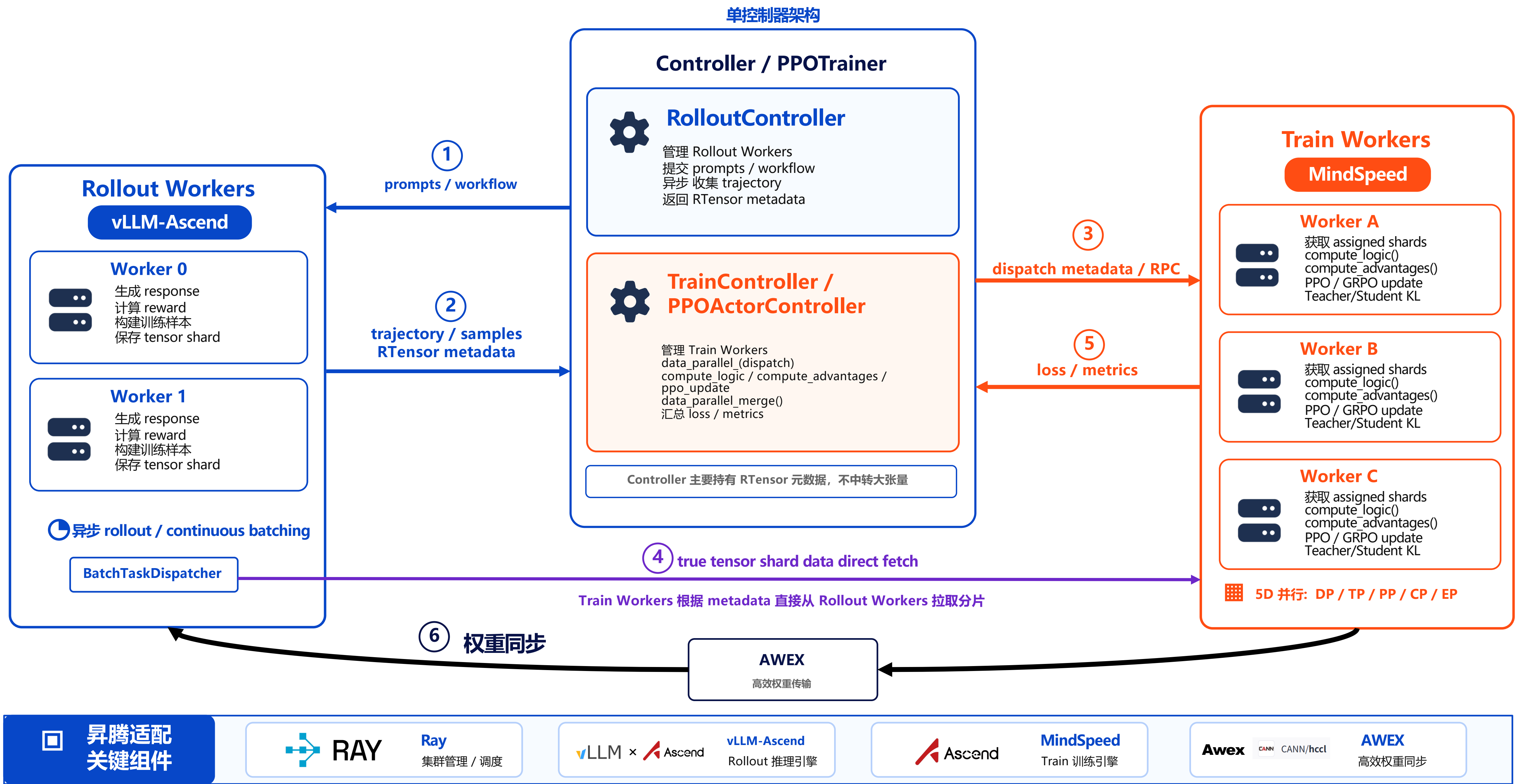


AREAL–DTA: Dynamic Tree Attention for Efficient Reinforcement Learning of Large Language Models

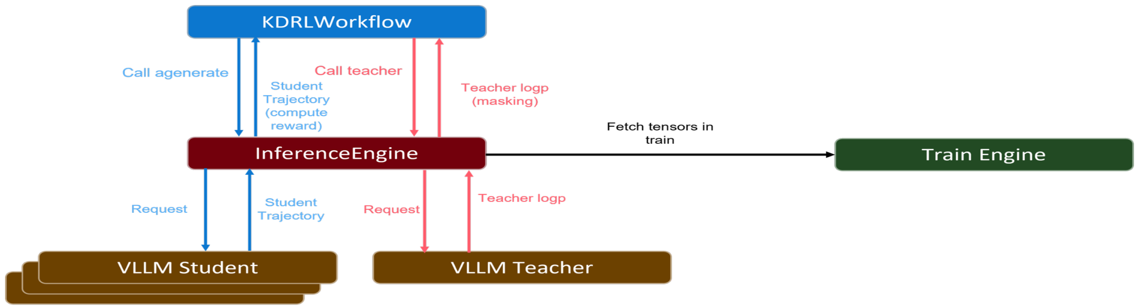
Up to 8.31× training throughput by not computing the same prefix twice

https://icml.cc/virtual/2026/poster/62287

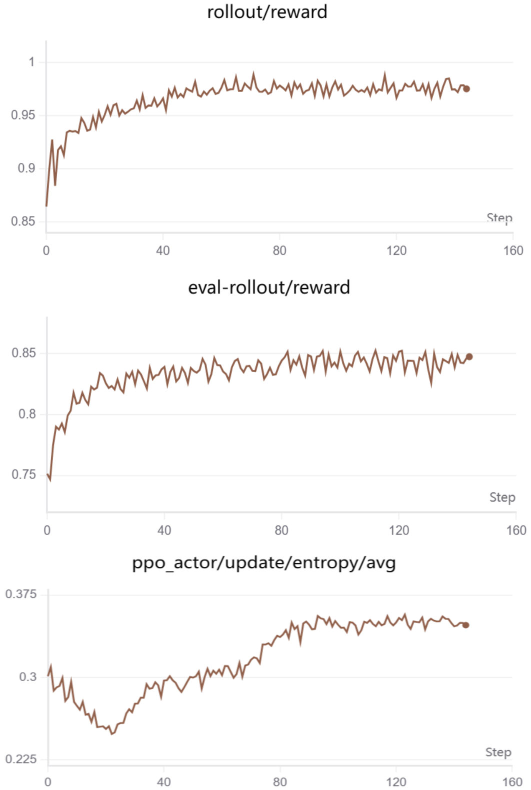
AREAL 2.0 – AREAL ON ASCEND



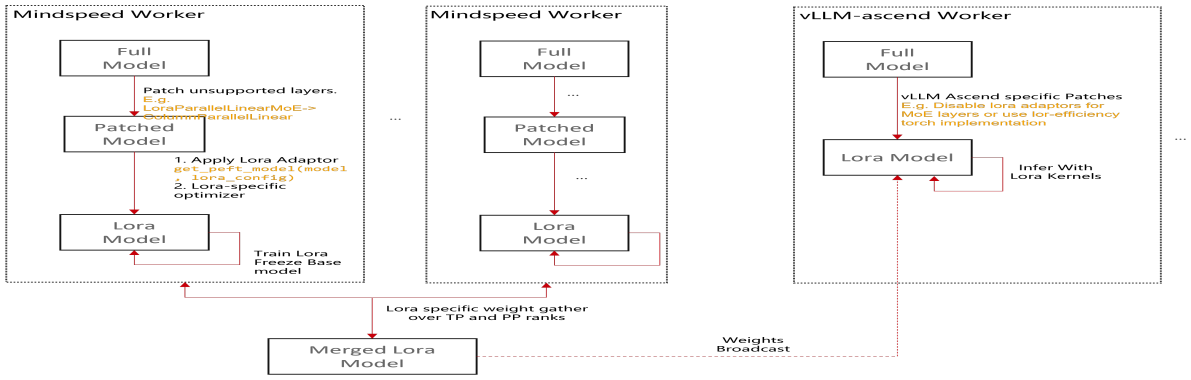
On Policy Distillation



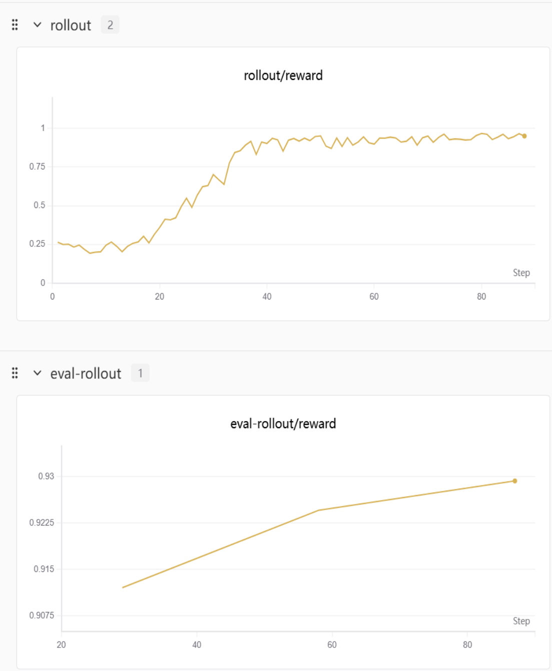
Setting	Value
Teacher	Qwen3-235B-A22B
Student	Qwen3-8B
Group size	16
Batch size	128
Teacher accuracy (eval)	82.35%
Teacher accuracy (train)	93.88%



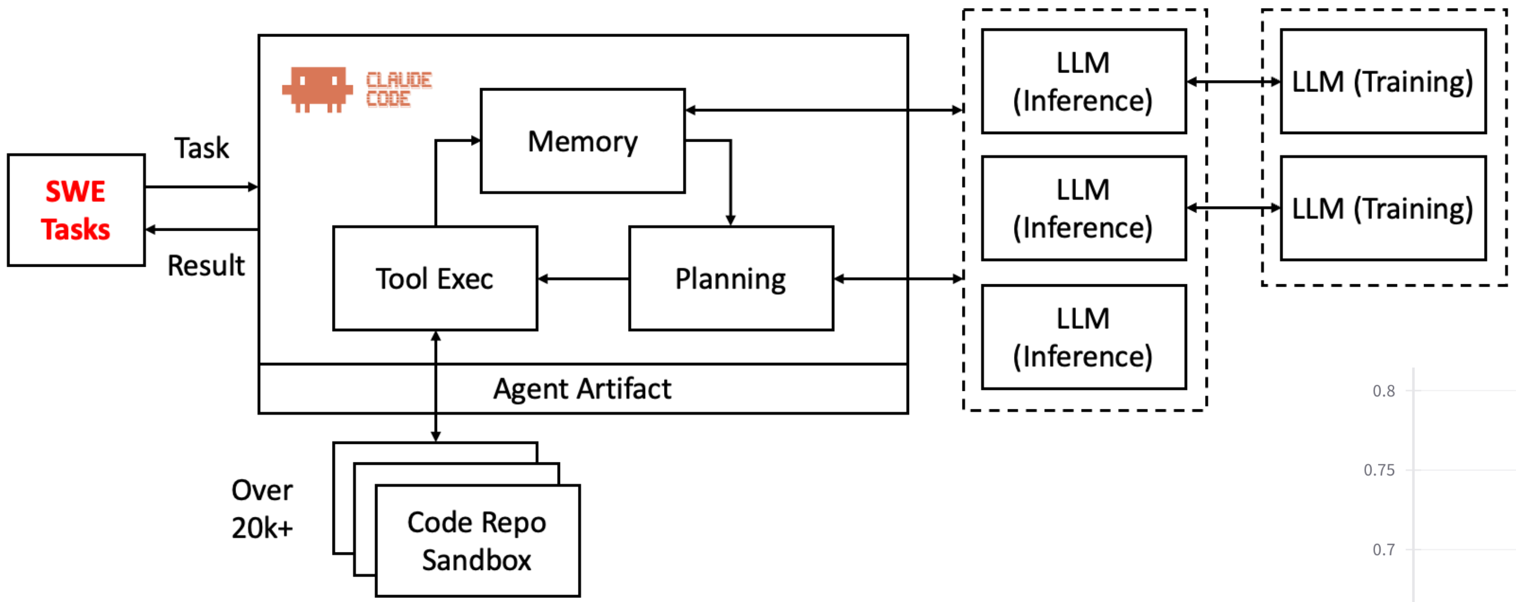
LoRA RL

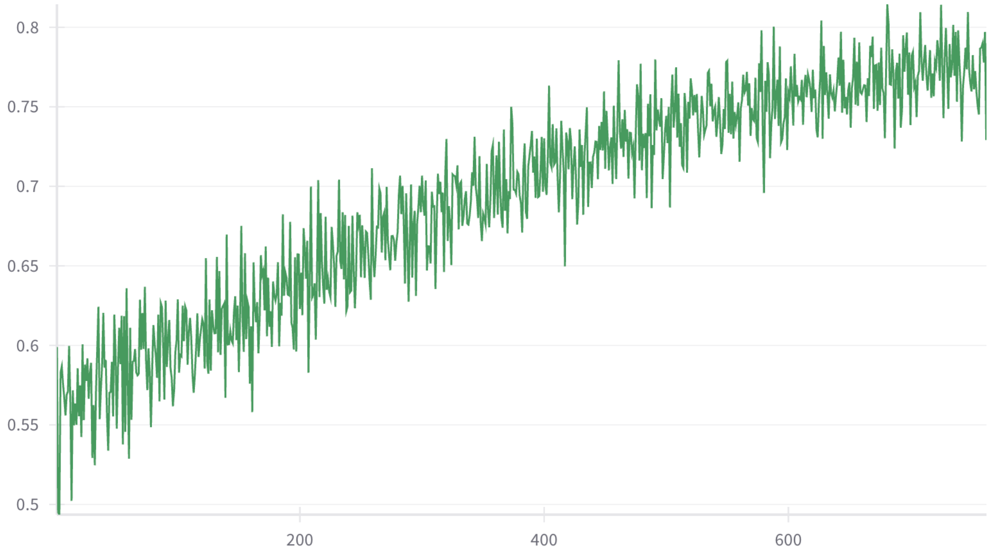


Param	Value
Model	Qwen3-30B-Base
Dataset	gsm8k
Nodes	2 (16 cards per node)
Trainer config	megatron:(attn:d1p4t4c1|ffn:d1p4t1e4)
Rollout Config	vllm:d1p4t4
Lora adaptors	qkv_linear, linear_proj
Lora Rank	128
Lora Alpha	128
LR	1e-5



Claude-Code Based RL Training Exp.





Biweekly Meeting 分享 SWE 训练经验

Features:

- Self-Evovling base on Memory + RL
- VLM training optimization and Omni RL
- Agent Memory System with long-horizon agent and cross-session sharing
- Global KVCache co-designed with agent memory system

Research:

- Online RL with self-rewarding from context
- 7 × 24 always-run Agent Performance

Summary

- RL infra is about ***connection***
- To help everyone build their own AI agents *easily and affordably*

Welcome to AReaL Community~

GitHub: https://github.com/areal-project/AReaL

开发者微信群:

个人微信: